

GraphQL - Security Overview and Testing Tips

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Content

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With the increasing popularity of GraphQL technology we are summarizing some documentation and tips about common security mistakes.

What is GraphQL?

[GraphQL](#) is a data query language developed by Facebook and publicly released in 2015. It is an alternative to REST API.

Even if you don't see any GraphQL out there, it is likely you're already using it since it's running on some big tech giants like Facebook, GitHub, Pinterest, Twitter, HackerOne and a [lot more](#).

A few key points on this technology

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GraphQL provides a complete and understandable description of the data in the API and gives clients the power to ask for exactly what they need. **Queries always return predictable results.**

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While typical REST APIs require loading from multiple URLs, GraphQL APIs get **all the data** your app needs **in a single request**.

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GraphQL APIs are organized in terms of types and fields, not endpoints. You can access the full capabilities of **all your data from a single endpoint**.

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GraphQL is **strongly typed** to ensure that application only ask for what's possible and provide clear and helpful errors.

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New fields and types can be added to the GraphQL API **without impacting existing queries**. **Aging fields can be deprecated** and hidden from tools.

Before we start diving into the GraphQL security landscape, here is a brief recap on how it works. The [official documentation](#) is well written and was really helpful.

A GraphQL query looks like this:

Basic GraphQL Query

```
query{
  user{
    id
    email
    firstName
    lastName
  }
}
```

While the response is JSON:

Basic GraphQL Response

```
{
  "data": {
    "user": {
      "id": "1",
      "email": "paolo@doyensec.com",
      "firstName": "Paolo",
      "lastName": "Stagno"
    }
  }
}
```

Security Testing Tips

Since [Burp Suite](#) does not understand GraphQL syntax well, I recommend using the [graphql-ide](#), an Electron based app that allows you to edit and send requests to a GraphQL endpoint; I also wrote a small python script [GraphQL_Introspection.py](#) that enumerates a GraphQL endpoint (with introspection) in order to pull out documentation. The script is useful for examining the GraphQL schema looking for information leakage, hidden data and fields that are not intended to be accessible.

The tool will generate a HTML report similar to the following:

GraphQL Schema

<http://192.168.1.82/examples/04-bank/graphql.php>

Legend

- Queries
- Mutations
- Arguments
- Types: String, Float, Int, null, [list]
- ~~Deprecated~~
- Fields

Available Operations Types:

- Query

- OBJECT
 - Query
 - user
 - Returns user by id (in range of 1-5)
 - User
 - id
 - IID
 - viewer
 - Represents currently logged-in user (for the sake of example - simply returns user with id == 1)
 - User
 - stories
 - Returns subset of stories posted for this Bank
 - [Story]
 - after
 - Fetch stories listed after the story with this ID
 - ID
 - 10
 - limit
 - Number of stories to be returned
 - Int
 - lastStoryPosted
 - Returns last story posted for this Bank
 - Story

[Introspection](#) is used to ask for a GraphQL schema for information about what queries, types and so on it supports.

As a pentester, I would recommend to look for requests issued to **“/graphql”** or **“/graphql.php”** since those are usual GraphQL endpoint names; you should also search for **“/graphiql”**, **“graphql/console/”**, online GraphQL IDEs to interact with the backend, and **“/graphql.php?debug=1”** (debugging mode with additional error reporting) since they may be left open by developers.

When testing an application, verify whether requests can be issued without the usual authorization token header:

```

POST /graphql HTTP/1.1
Host: dev.doyensec.com
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10.13; rv:59.0) Gecko/20100101 Firefox/59.0
Accept: */*
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
content-type: application/json
authorization: Bearer
eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjVhZTclMzQ1ODE1OTkwMDAwZmM0N2UwYyIsImVtYWlsIjoiZGQtdGVzdDRAZG95Z
W5zZWMuY29tIiwicm9sZSI6IiVTRVVIiLCJlbWFBbFZlcm1maWVkijp0cnVlLCJpYXQiOiE1MjU3NjY1NDgsImV4cCI6MTUyODM1ODU0OH0.P31
o4uLS22ZfYLUuU2O4f7hXLkCSG4psYtMCgD-2WeaSEz8Jk_AcOkH2i0Oc4XLE2JLTLN7VvFfZFrxa2u2pGLSzV7iob0JpBuogfUhVtzh5y1Yp
10lm6i6d9hKYuXltY7U9hd06ESHNha7xh8ixnpqSPCd3q0PpUvLzYDc09U2DFF19JOS2EZnamiRnA_UZKSYbYyG1RvzXCQNso04oXYYLH5N06
Gg0pMbpz_CLNBKYhyOIxsjYw71s_sRBOM2EaYdxplnngRa5Ufn2prwHZGQqph0UyW21oiukd2X4zAEkcpV1fBZ0Cg_1x5JYwAwqtxsK_U5p0g1
26DQtcmLA
Content-Length: 158
Connection: close

{"operationName":"RenewTwoFactor","variables":{},"query":"query RenewTwoFactor {\n  renewTwoFactor {\n    uri\n    secret\n    __typename\n  }\n}\n"}

```

Since the GraphQL framework does not provide any means for securing your data, developers are in charge of implementing access control as stated in the documentation:

“However, for a production codebase, delegate authorization logic to the business logic layer”.

Things may go wrong, thus it is important to verify whether a user without proper authentication and/or authorization can request the whole underlying database from the server.

When building an application with GraphQL, developers have to map data to queries in their chosen database technology. This is where security vulnerabilities can be easily introduced, leading to **Broken Access Controls**, **Insecure Direct Object References** and even **SQL/NoSQL Injections**.

As an example of a broken implementation, the following request/response demonstrates that we can fetch data for any users of the platform (cycling through the ID parameter), while simultaneously dumping password hashes:

Query

```

query{
  user(id: 165274){
    id
    email
    firstName
    lastName
    password
  }
}

```

Response

```

{
  "data": {
    "user": {
      "id": "165274",
      "email": "johndoe@mail.com",
      "firstName": "John",
      "lastName": "Doe",
      "password": "5F4DCC3B5AA765D61D8327DEB882CF99"
    }
  }
}

```

Another thing that you will have to check is related to information disclosure when trying to perform illegal queries:

Information Disclosure

```

{
  "errors": [
    {
      "message": "Invalid ID.",
      "locations": [
        {
          "line": 2,
          "column": 12
        }
      ],
      "Stack": "Error: invalid ID\n at\n (/var/www/examples/04-bank/graphql.php)\n"
    }
  ]
}

```

Even though GraphQL is **strongly typed**, SQL/NoSQL Injections are still possible since **GraphQL is just a layer between client apps and the database**. The problem may reside in the layer developed to fetch variables from GraphQL queries in order to interrogate the database; variables that are not properly sanitized lead to old simple SQL Injection. In case of MongoDB, NoSQL injection may not be that simple since we cannot “juggle” types (e.g. turning a string into an array. See [PHP MongoDB Injection](#)).

GraphQL SQL Injection

```
mutation search($filters Filters!){
  authors(filter: $filters)
  viewer{
    id
    email
    firstName
    lastName
  }
}

{
  "filters":{
    "username":"paolo' or 1=1--"
    "minstories":0
  }
}
```

Beware of nested queries! They can allow a malicious client to perform a DoS (Denial of Service) attack via overly complex queries that will consume all the resources of the server:

Nested Query

```

query {
  stories{
    title
    body
    comments{
      comment
      author{
        comments{
          author{
            comments{
              comment
              author{
                comments{
                  comment
                  author{
                    comments{
                      comment
                      author{
                        name
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}

```

An easy remediation against DoS could be setting a timeout, a maximum depth or a query complexity threshold value.

Keep in mind that in the [PHP GraphQL implementation](#):

-

Complexity analysis is disabled by default

-

Limiting Query Depth is disabled by default

-

Introspection is enabled by default. It means that anybody can get a full description of your schema by sending a special query containing meta fields **type** and **schema**

Outro

GraphQL is a new interesting technology, which can be used to build secure applications. Since developers are in charge of implementing access control, applications are prone to classical web application vulnerabilities like **Broken Access Controls**, **Insecure Direct Object References**, **Cross Site Scripting (XSS)** and **Classic Injection Bugs**. As any technology, GraphQL-based applications may be prone to development implementation errors like this real-life [example](#):

“By using a script, an entire country’s (I tested with the US, the UK and Canada) possible number combinations can be run through these URLs, and if a number is associated with a Facebook account, it can then be associated with a name and further details (images, and so on).”

@voidsec

Resources:

- <https://en.wikipedia.org/wiki/GraphQL>
- <https://dev-blog.apollodata.com/the-concepts-of-graphql-bc68bd819be3>
- <https://graphql.org/learn/>
- <https://www.howtographql.com/>
- <https://www.hackerone.com/blog/the-30-thousand-dollar-gem-part-1>
- <https://hackerone.com/reports/291531>
- <https://labs.detectify.com/2018/03/14/graphql-abuse/>
- <https://medium.com/the-graphqlhub/graphql-and-authentication-b73aed34bbeb>
- <http://www.petecorey.com/blog/2017/06/12/graphql-nosql-injection-through-json-types/>
- <https://webonyx.github.io/graphql-php/>

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